

DISCOVER LATENT HABITS

Repeated behavioral evidence; no explicit preference declaration

Behavior Stream

tool events
parsed actions
timestamps

Evidence Aggregation

action-entity counts
recent summaries
windowed evidence

Habit Induction

Frequency Gate

evidence filter
count $\geq n_{\min}$

Rule-based Generalization

glob patterns
normalized keys

Schema LLM Induction

habit fields
+ consequence

Persistent Habit Store

action-entity trigger
context + consequence
confidence + frequency
recency + lifecycle
generalized variants

CONTROL WHEN TO INTERVENE

A matched habit is a candidate, not an automatic notification

current event

Generalized Habit Matching

exact action-entity match
glob / normalized variants
trace label: variant_match

Selective Checks

Phase Gate (Code)
Think (ambiguous)
Code cases

Intervention Policy

Fixed score
+ threshold
Per-habit cooldown

Structured
PUSH

OPTIONAL

Natural-
language
realization
content only

Ambient Memory
agent context

STRUCTURED NO-PUSH / ABSTAIN

no habit match | phase/Think suppression | per-habit cooldown

Audit trace: habit match, phase/Think result, cooldown state, and final outcome

Induced habits

EVALUATED